

Atividade 03 - Método Simplex (Parte 2)

Dom Seg Ter Qua Qui Sex Sáb

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1- $\max Z = 4x_1 + 3x_2$

sujeito a $x_1 + x_2 \leq 5$ $x_1 + x_2 + s_1 = 5$

$2x_1 + x_2 \leq 8$ $2x_1 + x_2 + s_2 = 8$

$x_1, x_2 \geq 0$

SBV $\Rightarrow (x_1, x_2, s_1, s_2) = (0, 0, 5, 8)$

Base	x_1	x_2	s_1	s_2	b
s_1	1	1	1	0	5 $5/1 = 5$
s_2	2	1	0	1	8 $8/2 = 4 \checkmark$ $L_2 = L_2/2$
Z	-4	-3	0	0	0

↗

Linha	x_1	x_2	s_1	s_2	b
L_1	1	1	1	0	5 $L_1 = L_1 - L_2$
L_2	1	$1/2$	0	$1/2$	4 $L_2 = L_2 + 4L_2$
L_2	-4	-3	0	0	0

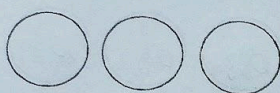
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Base	x_1	x_2	s_1	s_2	b
s_1	0	$1/2$	1	$-1/2$	1 $1/(1/2) = 2 \checkmark$ $L_1 = L_1 \cdot 2$
x_1	1	$1/2$	0	$1/2$	4 $4/(1/2) = 8$
Z	0	-1	0	2	16

↗

Linha	x_1	x_2	s_1	s_2	b
L_1	0	1	2	-1	2 $L_2 = L_2 - 1/2 L_1$
L_2	1	$1/2$	0	$1/2$	4 $L_2 = L_2 + L_1$
L_2	0	-1	0	2	16





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Base	x_1	x_2	s_1	s_2	b	$x_1=3, x_2=2, Z=18$
x_2	0	1	1	-1	2	
x_1	1	0	-1	1	3	
Z	0	0	2	1	18	

Conferência

$$3+2 \leq 5, 2(3)+2 \leq 8 \text{ e } Z=4(3)+3(2)=18 \checkmark$$

$$2. \max Z = 40x_1 + 50x_2$$

$$\text{sujeito a } \begin{aligned} 2x_1 + x_2 &\leq 100 & 2x_1 + x_2 + s_1 &= 100 \\ x_1 + 3x_2 &\leq 90 & x_1 + 3x_2 + s_2 &= 90 \\ x_1, x_2 &\geq 0 \end{aligned}$$

$$SBV \Rightarrow (x_1, x_2, s_1, s_2) = (0, 0, 100, 90)$$

Base	x_1	x_2	s_1	s_2	b	
s_1	2	1	1	0	100	$100/1=100$
s_2	1	3	0	1	90	$90/3=30 \quad L_2=L_2/3$
Z	-40	-50	0	0	0	

Linha	x_1	x_2	s_1	s_2	b	
L_1	2	1	1	0	100	$L_1=L_1-L_2$
L_2	$\frac{1}{3}$	1	0	$\frac{1}{3}$	30	
L_2	-40	-50	0	0	0	$L_2=L_2+50L_2$



$$\frac{-100}{3} + \frac{50}{3} = \frac{-20}{3}$$

Base	x_1	x_2	s_1	s_2	b	
s_1	$5/3$	0	1	$-1/3$	20	$20 \cdot \frac{3}{5} = 12 \checkmark$ $L_1 = L_1/5 = 3L_1$
x_2	$2/3$	1	0	$1/3$	30	$30 \cdot 3 = 90$
Z	$-20/3$	0	0	$50/3$	2500	

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Linha	x_1	x_2	s_1	s_2	b	
L_1	1	0	$3/5$	$-1/5$	12	
L_2	$2/3$	1	0	$1/3$	30	$L_2 = L_2 - \frac{2}{3}L_1$
L_3	$-20/3$	0	0	$50/3$	2500	$L_3 = L_3 + \frac{20}{3}L_1$

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Base	x_1	x_2	s_1	s_2	b	
x_1	1	0	$3/5$	$-1/5$	12	$\frac{1}{3} + \frac{1}{3} = \frac{2}{3} + 5 = \frac{17}{3} \div 3 = 2$
x_2	0	1	$-2/5$	$2/5$	16	$\frac{4}{3} + 30 = \frac{-42 + 90}{3} = \frac{48}{3} = 16$
Z	0	0	14	12	2480	$\frac{3}{3} \quad \frac{3}{3} \quad \frac{3}{3}$

$$\frac{70}{3} \cdot \frac{3}{5} = \frac{70}{5} \quad \frac{70}{3} \cdot \left(-\frac{1}{5}\right) = -\frac{70}{15} + \frac{50}{3} = -\frac{70}{15} + \frac{250}{15} = \frac{180}{15} = 12$$

$$\frac{40}{3} \cdot \frac{3}{5} = 14 \cdot 70 = 980$$

$$x_1 = 12, x_2 = 16, Z = 2480$$

Conferência

$$2(12) + 16 \leq 200, 12 + 3(16) \leq 90, 4$$

$$Z = 40(12) + 50(16) = 2480 \checkmark$$

